COURSE 2 · ALGEBRA · MODULE 2.3

Rearranging Formulas

Theory reader, audio edition

Changing the subject of a formula: the single most useful algebra skill for first-year physics, chemistry and biology.

Listening time about 50 minutes. Equations are spoken in words; numbered equations appear in conventional notation on the Visual Companion's Symbols Sheet.

Before you start

How to use this reader

This is the audio edition of the theory reader for Module 2.3. It is written to be listened to in a text-to-speech app as comfortably as it can be read on the page. Every equation is spoken in words, and every step of every worked example is narrated, so you never need to look at a symbol to follow the argument.

Each important equation carries a reference number, like this one: [Equation 1]. The same number appears on the **Symbols Sheet**, the first page of the Visual Companion, where the equation is printed in conventional notation. A good routine is to listen once on a walk, then read the Symbols Sheet with a pencil before opening the workbook.

Listening time is about fifty minutes. The module as a whole, including the workbook and the interactive explorer, is designed for one week at four hours.

What you should be able to do by the end

One. Rearrange any formula you meet in first-year physics, chemistry or biology to make a chosen quantity the subject.

Two. Handle the four hard cases: roots and powers, reciprocals, a target that appears more than once, and exponentials.

Three. Check a rearrangement four ways: by substitution, by units, by limiting cases and by scaling.

Four. Turn a curved law into a straight-line graph, which is how experimental scientists extract constants from data.

Section 1

Why this module comes first

Almost every calculation in the natural sciences begins the same way. A textbook gives you a law with one quantity on the left and several on the right, and the question asks for one of the quantities on the right.

The pendulum law gives the period from the length. The experiment measures the period and wants the gravitational field strength. The decay law gives the amount of material remaining after a time. The archaeologist knows the amount remaining and wants the time. The equation is the same in each case; what changes is which quantity you are hunting for. Rearranging is the skill of turning the equation around so that the quantity you want stands alone.

Students who struggle in first-year physics and chemistry rarely struggle with the physics or the chemistry. They struggle here. They can follow a lecture about energy conservation, but when a problem requires them to make velocity the subject of the kinetic energy formula, they stall, or they guess, or they substitute numbers too early and drown in a calculator. This module removes that bottleneck before it can form.

You bring an advantage. Anyone who has worked with a regression model has already done this. If an outcome equals an intercept plus a coefficient times a predictor, and you have asked what value of the predictor would produce a given outcome, you have rearranged a linear equation. You subtracted the intercept and divided by the coefficient. Everything in this module is that move, generalised and made reliable.

Section 2

An equation is a claim of balance

An equation says that two quantities are equal. The left side and the right side are two descriptions of the same number. That is the entire foundation of rearrangement, and it gives one rule: **whatever you do to one side, you must do to the other.**

The traditional picture is a beam balance with a pan on each side. If the pans balance, they still balance after you add a kilogram to both, or remove a kilogram from both, or double the contents of both, or take the square root of both, provided both pans hold positive amounts. What you may not do is change one pan alone.

This sounds childishly obvious, and it is. But almost every rearrangement error is a violation of it, usually a subtle one. The classic example is applying an operation to one term instead of to a whole side. If the left side is one over f and the right side is one over u plus one over v, you may take the reciprocal of each whole side. You may not take the reciprocal of each term separately. We will return to this trap in Section 7, because it is the most common error in first-year optics.

> **A useful habit**
>
> When you perform a step, say it aloud as an instruction to both sides. Not 'move the three across', but 'subtract three from both sides'. The language of moving terms across the equals sign is a shortcut that works until it doesn't. The language of doing the same thing to both sides never fails.

Section 3

Inverse operations: the undo list

To get a quantity on its own, you strip away everything that has been done to it. Each operation has an inverse that undoes it. There are five pairs worth knowing by heart, and they are printed as a table on the Symbols Sheet.

Addition and subtraction. Adding five is undone by subtracting five.

Multiplication and division. Multiplying by m is undone by dividing by m, provided m is not zero.

Powers and roots. Squaring is undone by taking the square root; cubing by the cube root. More generally, raising to the power n is undone by raising to the power one over n. Square roots need care with signs, which Section 10 covers.

Exponentials and logarithms. Raising e to a power is undone by the natural logarithm, written l n. Raising ten to a power is undone by log base ten. A logarithm simply answers the question: what power produced this number?

Trigonometric functions and their inverses. Taking the sine of an angle is undone by the inverse sine, also called arcsine. These appear in Course 1 and in the physics of waves.

Two operations are their own inverses and deserve a mention. Taking a reciprocal, that is, one over something, is undone by taking the reciprocal again. Changing sign, multiplying by minus one, is undone by changing sign again.

Section 4

The order of undoing

Knowing the inverses is not enough. You also need to apply them in the right order, and the right order is the reverse of the order in which the formula was built.

Think of getting dressed. You put on socks, then shoes. To undress, you take off shoes, then socks. Reverse order. The same is true of any quantity buried in a formula: the last thing done to it is the first thing you undo.

To find that order, imagine evaluating the formula with a calculator, starting from the quantity you are hunting for. Take the temperature conversion: Fahrenheit equals nine-fifths of the Celsius temperature, plus thirty-two. Starting from a Celsius value, your calculator would first multiply by nine-fifths, and then add thirty-two. So to undo it, you first subtract thirty-two, then multiply by five-ninths. The Celsius temperature equals five-ninths of the quantity Fahrenheit minus thirty-two.

The Visual Companion draws this as a **build tree**, a small diagram that shows the target at the bottom and each operation wrapped around it like layers of an onion. Rearranging is peeling the onion from the outside in. This picture is worth internalising, because it turns rearrangement from a bag of tricks into a single procedure: identify the outermost layer, apply its inverse to both sides, and repeat.

Section 5

Worked example: a straight line in disguise

The first equation of motion says that final velocity equals initial velocity plus acceleration multiplied by time. In symbols, v equals u plus a t. [Equation 1]

Suppose we want the acceleration. Build the tree: start with a, multiply by t, then add u. Two layers. The outer layer is the addition of u, so we remove it first.

Step one. Subtract u from both sides. The right side becomes a times t. The left side becomes v minus u. So v minus u equals a t.

Step two. The remaining layer is multiplication by t. Divide both sides by t. Acceleration equals v minus u, all divided by t. [Equation 2]

Notice the phrase 'all divided by'. When you hear it, picture a fraction bar that runs under the whole expression v minus u. That spoken phrase replaces the brackets you would see on the page. It also gives the formula a meaning you can check: acceleration is the change in velocity divided by the time taken. A correct rearrangement usually says something sensible when you read it in plain language. An incorrect one usually says something absurd.

Section 6

Worked example: the pendulum

The period of a simple pendulum, the time for one full swing there and back, equals two pi times the square root of the length divided by the gravitational field strength. Spoken more carefully: T equals two pi, multiplied by the square root of the fraction L over g. [Equation 3]

This formula has a long history in measurement. Before satellites, pendulums were the standard instrument for measuring the strength of gravity at different places on Earth, because a period can be timed very precisely. So the useful version of this formula is not the one that gives T, but the one that gives g.

Build the tree for g. It sits in the denominator of a fraction, so the innermost operation is taking the reciprocal of g and multiplying by L. Then comes the square root. Then multiplication by two pi. Three layers, and we peel from the outside.

Step one. Divide both sides by two pi. Now T over two pi equals the square root of L over g.

Step two. Square both sides to remove the root. The left side becomes T squared over four pi squared, because squaring two pi gives four pi squared. The right side becomes L over g.

Step three. g is still in a denominator. Take the reciprocal of both sides, which is legal because each side is a single fraction. Four pi squared over T squared equals g over L.

Step four. Multiply both sides by L. The gravitational field strength equals four pi squared times L, all divided by T squared. [Equation 4]

If instead you wanted the length, for example to build a clock with a period of exactly two seconds, you would stop after step two and multiply by g: L equals g times T squared, all divided by four pi squared. [Equation 5] The workbook asks you to do this calculation, and the answer is just under one metre. Grandfather clocks are about that size for exactly this reason.

> **Rearrange first, substitute last**
>
> Resist the urge to put numbers in early. A symbolic answer can be checked, reused for a different set of numbers, and inspected for sense. A string of decimals can do none of those things. Professional scientists substitute at the very last line.

Section 7

Fractions and the reciprocal trap

The thin lens equation says that one over the focal length equals one over the object distance plus one over the image distance. One over f equals one over u plus one over v. [Equation 6]

The temptation is to flip every term and write f equals u plus v. This is wrong, and you can prove it is wrong with small numbers. One half plus one half equals one. But two plus two does not equal one. The reciprocal of a sum is not the sum of the reciprocals.

The correct method is to isolate the term containing the target, combine everything else into a single fraction, and only then take a reciprocal. Suppose we want v.

Step one. Subtract one over u from both sides. One over v equals one over f minus one over u.

Step two. Combine the right side into a single fraction. The common denominator is u times f. One over f becomes u over u f. One over u becomes f over u f. Their difference is u minus f, all over u f.

Step three. Now each side is a single fraction, so take the reciprocal of both. The image distance v equals u f, divided by the quantity u minus f. [Equation 7]

Electrical circuits use the same structure: the total resistance of resistors in parallel obeys a sum-of-reciprocals law. The same trap, and the same cure, applies there.

Section 8

When the target appears more than once

Everything so far assumed the target appears in exactly one place. When it appears in two or more places, peeling the onion stops working, because there are two onions. The cure is a four-move routine: clear fractions, expand, collect, factor.

Take the rational function y equals x plus two, all divided by x minus three. [Equation 8] Suppose we want x. It appears in the numerator and in the denominator.

Move one, clear the fraction. Multiply both sides by x minus three. y times the quantity x minus three equals x plus two.

Move two, expand. x y minus three y equals x plus two.

Move three, collect. Gather every term containing x on the left, and everything else on the right. Subtract x from both sides and add three y to both sides. x y minus x equals three y plus two.

Move four, factor. The left side has x in both terms, so pull it out: x times the quantity y minus one equals three y plus two. Now x appears once, and a single division finishes the job. x equals three y plus two, all divided by y minus one.

You have just found the inverse function. In statistics, this is the move you make when you convert a fitted logistic probability back into odds, or a transformed variable back to its original scale.

Watch for division by zero

The final step divided by y minus one, which is only allowed if y is not equal to one. That is not a technicality. The original function never takes the value one, however large x becomes. The algebra is quietly telling you about the shape of the graph.

Section 9

Exponentials and logarithms

Growth and decay processes, from radioactive nuclei to bacterial colonies to compound interest, follow exponential laws, in which the unknown sits in an exponent. To bring it down to the ground floor you need a logarithm.

Radioactive decay says that the number of undecayed nuclei equals the starting number, multiplied by e raised to the power of minus k t, where k is the decay constant and t is time. N equals N nought times e to the minus k t. [Equation 9] Suppose we want the time.

Step one. Isolate the exponential. Divide both sides by N nought. N over N nought equals e to the minus k t.

Step two. Take the natural logarithm of both sides. The natural log undoes e to the power, so the right side collapses to minus k t. The left side becomes the natural log of N over N nought.

Step three. Divide by minus k. A neat move absorbs the minus sign: the log of a fraction changes sign when you flip the fraction. So t equals the natural log of N nought over N, all divided by k. [Equation 10]

Compound growth works the same way, with any base instead of e. An amount A equals a principal P multiplied by one plus r, raised to the power t. [Equation 11] To find t, divide by P, take logs of both sides, and use the most important property of logarithms: the log of a power equals the exponent times the log of the base. The exponent t steps down in front: t times log of one plus r. Divide by that log to finish. Applied to a population growing at two per cent a year, this gives a doubling time of thirty-five years, which you may recognise from the rule of seventy.

The pH scale runs in the opposite direction. pH equals minus log base ten of the hydrogen-ion concentration. [Equation 12] To recover the concentration, multiply by minus one and raise ten to the power of each side. The concentration equals ten to the power minus pH. A pH of three means a concentration of one thousandth of a mole per litre.

Three log laws to keep in your pocket

The log of a product is the sum of the logs. The log of a quotient is the difference of the logs. The log of a power is the exponent times the log. Every rearrangement involving logarithms uses one or more of these.

Section 10

Squares, roots and signs

Squaring destroys information. Three squared is nine, and so is minus three squared. When you undo a square by taking a root, you must ask which of the two candidates the situation allows. The full answer to x squared equals nine is x equals plus or minus three.

In kinetic energy, which equals one half times mass times velocity squared, [Equation 16] rearranging for velocity gives the square root of two E over m, and we keep only the positive root because a speed cannot be negative. In other problems, such as a ball thrown upwards that returns past its starting height, both roots have physical meaning, and discarding one would lose a real answer.

Three further cautions apply. Never divide by a quantity that might be zero. Never take the logarithm of a negative number or of zero, because no real power produces them. And when you square both sides of an equation to remove a root, check your final answers in the original equation, because squaring can introduce solutions that were never there.

Section 11

Four ways to check your answer

A rearranged formula can be checked without looking up the answer. Learn these four tests, because in an exam or a laboratory nobody will hand you the solution.

Test one: substitution

Choose simple numbers for every quantity except the target, compute the target from your rearranged formula, and feed all the numbers back into the original. If both sides agree, the rearrangement is almost certainly right. The interactive workbook does exactly this behind the scenes: it checks your answer at random values rather than comparing text, so any correct form is accepted.

Test two: units

Every term in a valid equation has the same units. If your formula for acceleration produces metres per second rather than metres per second squared, you have made an error. This is called dimensional analysis, and Course 1, Module 2 treats it fully. For now, check that the units of your answer are the units of the thing you were hunting for.

Test three: limiting cases

Push one quantity to an extreme and ask whether the formula behaves sensibly. In the lens formula, image distance equals u f over u minus f. As the object moves very far away, u becomes huge, the f in the denominator becomes negligible, and the image distance approaches f. That is correct: distant objects focus at the focal point. If your formula had predicted the image racing off to infinity, you would know it was wrong.

Test four: scaling

Ask what happens if one input doubles. The pendulum period is proportional to the square root of length. So doubling the length should multiply the period by the square root of two, about one point

four one. Quadrupling the length should double the period. Scaling arguments are the physicist's favourite sanity check, and they are also how you answer 'by what factor' questions in seconds without any calculation.

Section 12

Linearisation: the experimentalist's use

Rearranging has a second purpose beyond solving for a number. It lets you turn a curved relationship into a straight line, and straight lines are what we can fit, judge and read constants from.

Return to the pendulum. A graph of period against length is a curve. But square both sides of the pendulum law and you get T squared equals four pi squared over g, multiplied by L. [Equation 13] Compare this with the equation of a straight line through the origin, y equals m x. If you plot T squared on the vertical axis and L on the horizontal, you get a straight line whose gradient is four pi squared over g. Fit the line, read off the gradient, and rearrange for g. That is how an undergraduate laboratory measures gravity.

Exponential decay linearises by taking logs. The natural log of N equals the natural log of N nought, minus k times t. [Equation 14] Plotting the log of the count against time gives a straight line with gradient minus k and intercept the log of N nought.

Power laws linearise on log–log axes. If y equals a times x to the power n, then log y equals log a plus n times log x. [Equation 15] The gradient of the log–log plot is the exponent n. Economists will recognise this as the constant-elasticity model, and you may well have estimated one. The procedure in a physics lab is the same procedure you have used in a regression package. Only the vocabulary differs.

> **Why straight lines?**
>
> The eye detects deviations from a straight line far better than deviations from a curve. Least-squares fitting of a line is simple and its uncertainties are well understood. And a straight line has only two parameters, each with a clear meaning. Linearisation converts a question about a law into a question about a gradient and an intercept.

Section 13

The procedure in seven steps

1. Identify the target. Circle it, or say its name aloud.

2. Count how many times it appears. If once, go to step three. If more than once, clear fractions, expand, collect and factor until it appears once.

3. Build the tree. Work out the order in which operations were applied to the target.

4. Peel from the outside. Apply the inverse of the outermost operation to both whole sides.

5. Repeat until the target stands alone.

6. Handle signs and domains. Choose between positive and negative roots, and note any values that would cause division by zero or the log of a non-positive number.

7. Check. Substitute numbers, test units, try a limiting case or a scaling.

Section 14

Spoken glossary

Subject of a formula. The quantity standing alone on one side. Rearranging is changing the subject.

Inverse operation. An operation that undoes another: subtraction undoes addition, a logarithm undoes an exponential.

Reciprocal. One divided by a quantity. The reciprocal of four is one quarter.

Factor out. Rewrite a sum of terms sharing a common factor as that factor times a bracket. x y minus x becomes x times y minus one.

Natural logarithm. The logarithm to base e, written l n. e is about two point seven one eight.

Domain. The set of values a quantity may take. Lengths have a positive domain; logarithms need positive inputs.

Linearise. Rearrange or transform a law so that a graph of it is a straight line.

Gradient. The slope of a straight line: rise divided by run.

Proportional to. Changes by the same factor. If y is proportional to x squared, tripling x multiplies y by nine.

Next: open the Visual Companion and work through the Symbols Sheet with a pencil, then try the Formula Untangler in the interactive explorer.